ESWARAN M

PCB DESIGN ENGINEER

+91 7358534319 | eswar28.m@gmail.com
LinkedIn: https://www.linkedin.com/in/eswar-m-82ab102b7/

PROFESSIONAL SUMMARY

PCB Design Engineer with 2+ years of hands-on experience in PCB layout and design, specializing in high-speed digital, RF, analog, power, and mixed-signal circuits. Experienced in designing multilayer PCBs up to 10 layers using Cadence Allegro and Altium Designer. Skilled in HDI technology, controlled impedance routing, differential pair routing, length matching, signal integrity, power integrity, and EMI/EMC considerations. Contributed to 50+ PCB designs across embedded, defence, automotive, medical, and industrial applications. Experienced in preparing manufacturing deliverables, including Gerber, ODB++, stack-up, and stencil files.

TECHNICAL SKILLS

PCB Design Tools: Cadence Allegro, OrCAD, Altium Designer (AD25)
PCB Technologies: Multilayer PCB Design, HDI, Rigid PCB, Flex PCB, Rigid-Flex PCB, Blind/Buried Vias, Via-in-Pad, ELIC
High-Speed Interfaces: DDR, PCIe, USB, Ethernet (RGMII/SGMII), LVDS, MIPI-DSI/CSI
PCB Design Expertise: Component Placement, Differential Pair Routing, Length Matching, Controlled Impedance Routing, Stack-Up Design, RF and Antenna Layout
Signal and Power Integrity: Signal Integrity (SI), Power Integrity (PI), EMI/EMC Mitigation
Manufacturing and Compliance: DFM, DFT, IPC Standards, Gerber Generation, ODB++, Stencil Files, Manufacturing Documentation

PROFESSIONAL EXPERIENCE

VVDN Technologies Pvt. Ltd. – Kochi | Engineer – PCB Design | July 2024 – Present

- Designed PCB layouts for analog, digital, RF, and power supply applications while following applicable IPC design standards.
- Executed multilayer PCB designs of up to 10 layers, incorporating high-speed interfaces such as DDR, PCIe, MIPI, USB, and Ethernet.
- Developed HDI PCB layouts using controlled impedance routing and blind/buried via structures.
- Performed component placement, differential pair routing, signal length matching, and routing optimization.
- Applied DFM/DFT guidelines and EMI/EMC design considerations to support manufacturability and design reliability.
- Prepared manufacturing deliverables, including Gerber, ODB++, stack-up, and stencil documentation.
- Collaborated with hardware, mechanical, simulation, and manufacturing teams to align design requirements and manufacturing constraints.

VVDN Technologies Pvt. Ltd. – Kochi | BOM Engineer | July 2022 – July 2024

- Reviewed Engineering Change Orders (ECOs) and updated Bill of Materials (BOM) records.
- Coordinated BOM management activities with engineering and cross-functional departments.
- Supported production planning and maintained ERP records.

SFO Technologies | BOM Engineer | May 2021 – July 2022

- Reviewed Engineering Change Orders (ECOs) and maintained updated BOM records.
- Coordinated BOM-related activities with engineering and other departments.
- Supported production planning, material coordination, and ERP maintenance.

KEY PCB DESIGN PROJECTS

Winch Board for Tidal Cam

- Designed a 10-layer PCB incorporating high-current power input, a central SoC, dual motor controllers, SD card storage, Ethernet, and sensor interfaces.
- Executed complete PCB layout, high-current routing, creepage/clearance planning, impedance-controlled Ethernet routing, and Gerber release for production.

M.2 Connector Board

- Designed a 10-layer mixed-signal PCB integrating FPGA, AGILUX, MIPI, PCIe interfaces, and multiple power domains.
- Executed component placement, MIPI signal length matching, power-clearance optimization, impedance-controlled routing, and Gerber release.

EDUCATION

Diploma in Electronics Engineering | 2014 – 2017
Srinivasan Polytechnic College, Perambalur | State Board of Technical Education